

Highlights

1. Human oversight pipelines saturate 2.71x faster than memoryless audit models predict
2. Review depth collapses non-linearly above utilization threshold $\rho_c = 0.60$
3. SEDI index measures governance observability from three public record streams
4. Artifact-count compliance dashboards cannot detect oversight saturation onset
5. Three governance-by-design principles treat oversight as a scaling constraint